INCH-POUND
MIL-DTL-15291/13C
21 January 2022
SUPERSEDING
MIL-DTL-15291/13B
6 February 2006

DETAIL SPECIFICATION SHEET

SWITCHES, ROTARY, SNAP ACTION CLASS 20SR BASE MOUNTED, SIDE CONNECTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-DTL-15291.

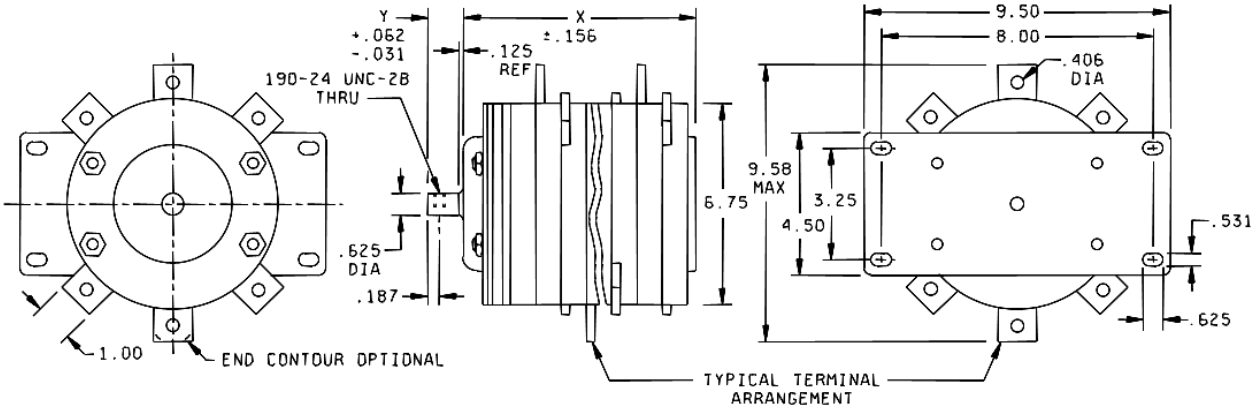
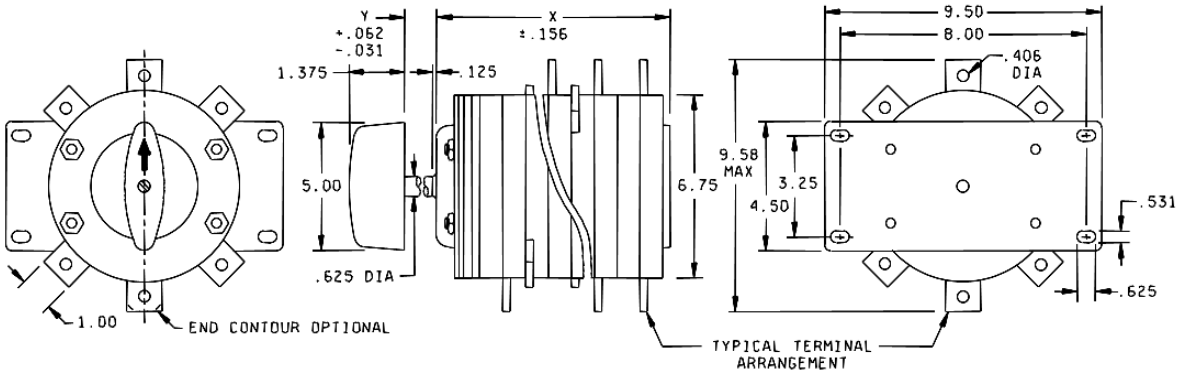


FIGURE 1 Class 20SR base mounted, side connected, with coupler shaft

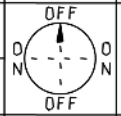
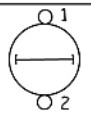
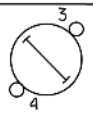
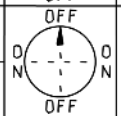
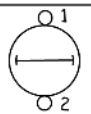
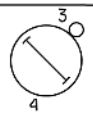
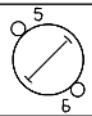
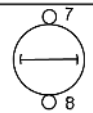
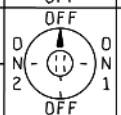
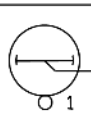
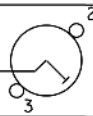
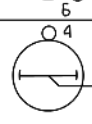
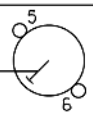
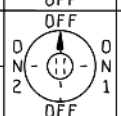
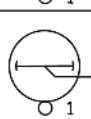
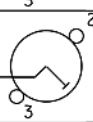
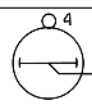
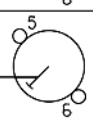
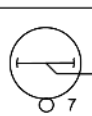
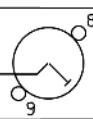
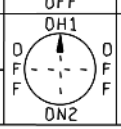
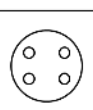
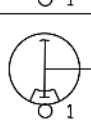
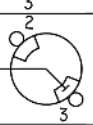
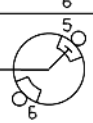
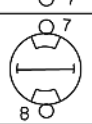
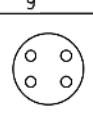
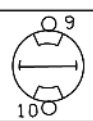


NOTES
1 Unless otherwise specified, tolerance is plus or minus 0.015.

FIGURE 2 Class 20SR base mounted, side connected with handle.

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TABLE I Type and switching characteristics.

M15291/13 DASH NO. TYPE DESIGNATION	DETAIL REF / TORQUE IN-LBS	DIM "X" / DIM "Y"	HANDLE OR SHAFT POSITION	CIRCUIT AND SPACER CONFIGURATIONS ROTOR POSITION, SPACER LOCATIONS, TERMINAL MARKING AND LOCATIONS NO 1	NO 2	NO 3	NO 4	NO 5	NO 6	NO 7	NO 8	NO 9	NOTES
-001	FIG 2	4.812											
	72	.437											
-002	FIG 2	6.343											
	72	.437											
-003	FIG 1	6.343											
20SR284	72	.750											
-004	FIG 1	8.093											
20SR384	72	.750											
-005	FIG 2	8.968											
20SR4S3	80	2											

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REQUIREMENTS:

Applicable specification: MIL-DTL-15291

Dimensions and mounting: See figures 1 and 2

Switching characteristics: See table I

Angle of throw: 90 degrees

Switching action: Snap action, reciprocating

Electrical and endurance ratings: See table II

Stop strength (applicable to switches with stops): 100 inch-pounds

Vibration: 50 Hz, MIL-STD-167-1

Shock (high impact): High impact, MIL-DTL-901

Contact resistance: 1.5 milliohms maximum

Dielectric withstanding voltage: 2,000 Vrms

Insulation resistance: 200 megohms minimum

Temperature rise: 50°C maximum

Mounting bolts: (4) ½ diameter, length, and headstyle to suit application. Bolts not furnished.

Handle: M15291/12-100 per MIL-DTL-15291/12

Terminal marking: Terminal markings shown in table I locate terminals as viewed from the front of the switch (handle end) Markings shall be stamped in rear surfaces of terminals.

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TABLE II Electrical and endurance ratings.

Tests	Current (amperes)	Voltage (volts)	Electrical Operations (number of operations)	Test rate (operations per minute)	De-energized operations (number of operations)
Alternating current (rms)	200	500	6,000	8	4,000
Direct current	200	250	4,000	3	--

GENERAL INFORMATION:

Switches not covered by specification sheets: Switches which are fabricated from standard parts, as used in qualified switches, but which do not comply with switches detailed herein with respect to circuit characteristics, switching action, mounting arrangement, and handle details may be acquired under this specification from contractors having qualification approval under this specification.

Extended ratings: Switches detailed herein have been tested and found satisfactory at the extended ratings listed in table III. Reduced life expectancy must be anticipated for switches used at these increased voltage or current levels. Tests under the conditions of table III are not required for qualification acceptance and they are not repeated routinely as for maintenance of qualification. Supplemental evaluations and tests applicable to particular circuit requirements are recommended.

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TABLE III. Extended ratings.

Switching characteristics	AC 60 or 400 Hz											
	125 volts				250 volts				500 volts			
	Resistive or lamp load		Inductive load 0.75 p.f.		Resistive or lamp load		Inductive load 0.75 p.f.		Resistive or lamp load		Inductive load 0.75 p.f.	
	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations
All	200	6,000	200	6,000	200	6,000	200	6,000	200	6,000	200	6,000
	DC											
	120 volts				250 volts				350 volts			
	Resistive or Lamp load		Inductive load[1]		Resistive or Lamp load		Inductive load[1]		Resistive or Lamp load		Inductive load[1]	
	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations
All	200	4,000	--	--	200	4,000	200	4,000	--	--	--	--

[1]0.025 henry for inductive circuit

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APPLICATION AND ACQUISITION GUIDE: PIN and type designation cross reference shall be as shown in table IV.

TABLE IV. Application and acquisition guide.

M15291/12 Dash No.	Type Designation	For new or existing design	For re-Placement	Circuit Configuration
-001	--	X		A (off-on-off-on)
-002	--	X		A (off-on-off-on)
-003	20SR2B4	X		B (off-on-1-off-on 2)
-004	20SR3B4	X		B (off-on-1-off-on 2)
-005	20SR4S3	X		S (on 1-on 2-off-off)

Referenced Documents:

MIL-DTL-15291
MIL-TL-15291/12
MIL-DTL-901
MIL-STD-167-1

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Navy - SH
DLA - CC

Preparing activity:
DLA - CC

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